UKUDLA Data Science Certificate

Remote Computing using SSH & Linux

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UKUDLA
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Supported by:
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With funding from the:
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With support from
Federal Ministry of Agriculture, Food and Regional Identity

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One project in different computing contexts

Learner’s laptop → Local Linux container → Remote Linux server → Shared compute infrastructure

Linux

- Operating-system family
- Common on servers and containers

Terminal

- Text-based interface
- Displays input and output

Shell

- Reads and runs commands
- Bash is a common shell

Why the command line matters in data science

Repeatable work

- Commands can be saved in scripts.
- The same instruction can process many files.
- Text commands are easy to document and review.

Portable work

- Remote systems may have no desktop.
- Small tools can be combined.
- Automation scales beyond manual clicking.

The command line is not only a different interface. It expresses repeatable operations in a form that computers and collaborators can execute.

SSH: one secure connection, different outcomes

GitHub repository access

- Clone, fetch, pull, and push
- Exchange Git objects
- No general-purpose shell

Remote computing

- Open a remote shell
- Run programs and manage files
- Transfer files securely

Public-key authentication

Private key
stays with the learner

cryptographic proof
private key is not transmitted

Public key
registered with a service

- The public key may be shared; the private key must remain private.
- A passphrase protects the private key if its file is exposed.
- The server checks that the client controls the matching private key.
- Host fingerprints help verify that the client reached the intended host.

A remote session changes the computing context

Local context	SSH connection	Remote context
local user	keyboard and text cross the network	remote user
local files		remote files
local processes		remote processes

Always establish context before acting:

- Which computer and user am I controlling?
- Which directory and repository am I in?
- Which files and processes will this command affect?

Navigate and combine small Linux tools

```
/
|-- home/
|   `-- student/
|       `-- analysis-project/
|-- tmp/
`-- work/
```

```
head results/summary.csv |
  column -s, -t
```

Navigation and composition

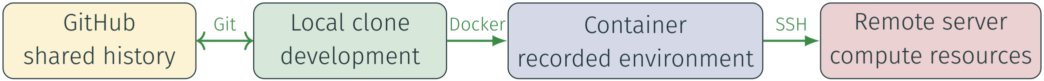
- Absolute paths begin at `/`.
- Relative paths begin at the current directory.
- A pipe connects one command's output to the next.

Commands operate in a context. Confirm the current directory before reading, writing, moving, or deleting files.

A container is a safe Linux practice environment

- The image supplies a consistent Linux environment.
- A temporary container can be explored and discarded.
- Source copied into the image is a build-time snapshot.
- Bind-mounted inputs and outputs remain on the host.

One data science project across several contexts



- Git identifies and exchanges project states.
- Docker describes and builds the execution environment.
- SSH provides secure access to another computer.
- Linux commands operate the project in containers and on servers.

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Command-line and remote-computing care

Before a command

- Confirm the host and user.
- Confirm the current directory.
- Inspect targets and command options.
- Understand persistence and permissions.

For shared infrastructure

- Protect credentials and private keys.
- Respect CPU, memory, and storage limits.
- Manage long-running work appropriately.
- Preserve outputs and stop unneeded processes.

SSH secures the connection. It does not make every command safe, every server trusted, or every analysis scientifically correct.

Key message: know which computer a command affects

1. Linux, a terminal, a shell, and SSH are related but distinct.
2. SSH authenticates and protects communication with a remote service.
3. A remote shell operates another computer, not the local laptop.
4. Linux commands provide a repeatable interface across containers and servers.
5. Git, Docker, SSH, and Linux connect one data science project across several computing contexts.

Detailed commands, exercises, verification, and troubleshooting are provided in the learning-module guides.

Supported by

South African government and the National Research Foundation

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Project Manager:
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